

		the balance length QJ' is shorter.	
3	A binary star system shown in Fig.3.1 consists of two stars orbiting around their common centre of mass, P. One such binary star system is known as Sirius where the mass of Sirius A is twice the mass of Sirius B. The mass of Sirius B is 1.99×10^{30} kg. Sirius A is at a distance of $\frac{1}{3}D$ from P. Sirius B is at a distance of $\frac{2}{3}D$ from P. Fig. 3.1		
	(a)	State <i>Newton's law of gravitation</i>. [2]	
		The mutual gravitational force of attraction between two <u>point</u> masses is <u>(directly) proportional to the product of their masses</u> and <u>inversely proportional to the square of their distance apart</u>.	
	(b)	The orbital period of this binary star system is 50 years. Determine the distance D between the binary stars, assuming that the stars move in circular orbits.	
		distance $D = \dots\dots\dots$ m	[4]
		The centripetal force is provided by the gravitational force. Consider Sirius B: $F_g = F_c$	

		$\frac{GM(2M)}{D^2} = M\omega^2 \left(\frac{2D}{3}\right)$ $\frac{GM(2M)}{D^2} = M \left(\frac{2\pi}{T}\right)^2 \left(\frac{2D}{3}\right)$ $D^3 = 3GMT^2/4\pi^2$ $= 3 \times 6.67 \times 10^{-11} \times 1.99 \times 10^{30} \times (365 \times 50 \times 24 \times 60 \times 60)^2 / 4\pi^2$ $D = 2.93 \times 10^{12} \text{ m}$ <i>Solution may also be obtained considering Sirius A.</i>	
	(c)	Determine the speed of a probe, originally at rest at infinity, when it reaches the centre of mass of the star system.	
		speed =	 m s ⁻¹ [3]
	L2	By principle of conservation of energy, Gain in kinetic energy = Loss in gravitational potential energy $KE_f - KE_i = GPE_i - GPE_f$ $\frac{1}{2}mv^2 - 0 = 0 - \left[G\left(\frac{M_A m}{R_A}\right) + G\left(\frac{M_B m}{R_B}\right) \right]$ $v = \sqrt{\frac{4 \times 6.67 \times 10^{-11} \times 1.99 \times 10^{30}}{\frac{1}{3}(2.93 \times 10^{12})} + \frac{6.67 \times 10^{-11} \times 1.99 \times 10^{30}}{\frac{2}{3}(2.93 \times 10^{12})}}$ $= \sqrt{\frac{3(6.67 \times 10^{-11} \times 1.99 \times 10^{30})}{(2.93 \times 10^{12})} \times \left(4 + \frac{1}{2}\right)}$ $= 26100 \text{ m s}^{-1}$	